12 V fan switched by 3.3 V logic — a gate booster built from my own parts

Arduino Pro Mini (3.3 V) pin D9 = OC1A. Q2 (2N7002) pulls the IRF7416 gate low; the IRF7416 then drives the IRFZ44N gate from +12 V.

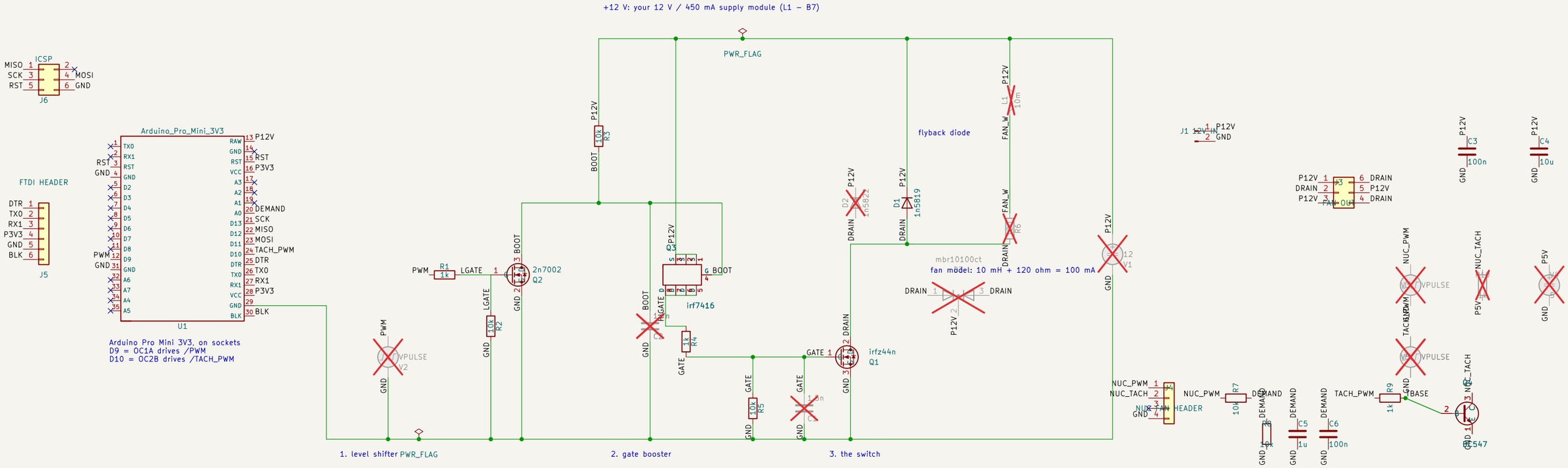
WHY THREE TRANSISTORS
The bins hold 12x IRFZ44N, 11x 2N7002 and 6x IRF7416, and no logic-level N-channel that can carry a 12 V / 100 mA fan:
- 2N7002: Id max 115 mA in my own spec record, and at Vgs = 3.0 V it only conducts 52 mA. As the pass element it starves the fan and burns 300 mW, over its 200 mW rating. Here it is only a level shifter, never the pass transistor.
- IRFZ44N: Rds(on) is specified only at Vgs = 10 V, and Vgs(th) is guaranteed 2.0 V .. 4.0 V. Wired straight to the pin, /GATE only reaches 3.0 V (R1/R5 divider): a typical unit runs the fan but wastes 5.2 mW, and a worst-case unit (Vgs(th) = 4.0 V) never turns on at all. sim/variants.py measures every case.

So the 3.3 V pin only drives small signals; the +12 V rail does the real gate drive.

MEASURED BY sim/run.sh (ngspice)
fan starts conducting at 2.27 V on the pin
fan current 102 mA, V(drain) 1.6 mV
turn-off spike clamped to 12.37 V
/GATE rise 5 tau = 7.5 us (R4 x C1)
/GATE settles at 10.91 V
Rds(on) 15.7 mOhm -> 0.16 mW in Q1
pin current never above 0.30 mA
/BOOT rise 5 tau = 85 us (R3 x C2)
fan L/R = 83 us, so at 1 kHz it is chopped 0..100 mA; average at 50 % duty = 54 mA

3 FANS ON ONE OUTPUT (sim/fan_count.py)
302 mA total, 4.8 mV at the drain, 1.5 mW in Q1, 295 mA peaks through the 1N5819 at turn-off (a 1 A part), 141 mA average from the 12 V module. Four fans also fit; four 150 mA fans would not - they need 452 mA and the L1 - B7 module only gives 450 mA.

HOW IT WORKS
1. Pin low: Q2 is off, no current flows in R3, so /BOOT is pulled to +12 V. Q3 (source on +12 V) sees Vgs = 0 and stays off; R5 holds /GATE at 0 V. The fan is off.
2. Pin high: R1/R2 give Q2's gate 3.0 V, Q2 turns on and sinks 12 V / 10 k = 1.2 mA, so /BOOT collapses to 12 mV. Q3 now sees Vgs = -12 V and turns on hard.
3. +12 V reaches Q1's gate through R4; R4 and R5 form a divider, so /GATE settles at 12 V x 10 k / 11 k = 10.9 V - the region where the datasheet quotes 17.5 mOhm.
4. Q1 is fully enhanced: Rds(on) 16 mOhm, V(drain) 1.6 mV, 0.16 mW lost, and the fan gets the whole 12 V. The Arduino pin itself never sources more than 0.3 mA.
5. The logic is not inverted, and at power-up the pin is floating: R2 keeps Q2 off, so Q3 is off, so Q1 is off. The fan cannot drift on.
6. Q3's SOURCE must be the +12 V side. Swap source and drain and the P-FET's body diode feeds +12 V into Q1's gate, and the fan can never switch off.
7. D1 (1N5819, cathode to +12 V) catches the fan winding's current when Q1 opens: /DRAIN peaks at 12.37 V instead of spiking hundreds of volts.
8. C1 and C2 are not purchased parts: they are the two MOSFETs' own input capacitances (Ciss) drawn explicitly so the gate RC time constants are visible. L1 + R6 are a model of the fan - the bins hold no 12 V fan.



TRY IT YOURSELF (Simulator > Run Simulator, then Transient or DC Sweep)
- Run the DC sweep: /BOOT falls from 12 V to 0.01 V while /GATE jumps from 0 to 10.9 V.
- Delete Q2, Q3, R3, R4 and C2 and wire R1 straight to /GATE: the fan still runs (100 mA), but Q1 now burns 5.2 mW instead of 0.16 mW. Then swap in irfz44n_wc.lib: fan stops dead.
- Shorten R3 to 1 k and watch the turn-off delay shrink from ~55 us to a few us.
- Remove D1 and look at the /DRAIN spike at turn-off: that is what the diode absorbs.
- Change V2's PULSE width tw from 500u to 250u (25 % duty): average fan current falls, the pulses the fan sees stay a full 12 V. Speed control without dimming the motor.

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* Simulation setup: KiCad hands these lines straight to ngspice.
* DC sweep of the pin drive 0 -> 3.3 V, then 1 kHz PWM (50 %) for 5 ms.
.op
.dc V2 0 3.3 0.01
.tran 1u 5m
.print dc v(/BOOT) v(/GATE) v(/DRAIN) i(v1) i(v2)
.print tran v(/GATE) v(/DRAIN) i(l1) i(v2)
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PARTS - all from my cabinets
U1 Arduino Pro Mini 3V3 L1 - C5 x1 (on three female sockets)
Q1 IRFZ44N TO-220 R1 - B4 x12
Q2 2N7002 SOT-23 R1 - A4 x11
Q3 IRF7416 SO-8 R1 - A4 x6 (a breakout board in the same drawer)
D1 1N5819 40 V 1 A Schottky L1 - E1 x30
D2 1N5822 40 V 3 A DO-27 L1 - E1 x9 spare, not soldered
D3 MBR10100CT 10 A dual L1 - E1 x3 spare, not soldered
Q4 BC547 TO-92 R1 - D3 x20
R1, R4 1 k R1 - A2 x20
R2, R3, R5 10 k R1 - B2 x20
V1 12 V supply module L1 - B7 x2
C1, C2, L1, R6 are model artefacts, not bin parts

PHYSICAL PINOUTS - symbols use package pin order: sheet numbers = board legs
Q2 2N7002 SOT-23 1 gate /LGATE 2 source /GND 3 drain /BOOT
Q3 IRF7416 SO-8 1,2,3 source /P12V 4 gate /BOOT 5..8 drain /HIGATE
Q1 IRFZ44N TO-220 1 gate /GATE 2 drain /DRAIN 3 source /GND
Q4 BC547 TO-92 1 emitter /GND 2 base /TBASE 3 collector /NUC_TACH
D1 1N5819 DO-41 banded end (cathode) to +12 V, other end to /DRAIN
J3 3 fans odd pins = +12 V, even pins = the switched /DRAIN side
ngspice's own order (D-G-S-B, C-B-E) lives in the sim/models/ subckt wrappers
J5 FTDI 1 DTR 2 TXO 3 RX1 4 P3V3 5 GND 6 BLK
J6 ICSP 1 MISO 3 SCK 4 MOSI 5 RST 6 GND (pin 2 stays dead)
U1 35 module holes: symbol pin number = socket pad number
docs/pcb-notes.md carries the full pad-to-net table for the board.

klipper-cura - parts from cabinets L1 and R1

Sheet: /
File: mosfet-fan-switch.kicad_sch

Title: 12 V fan switch from 3.3 V logic - gate booster

Size: A3	Date: 2026-10-21	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/1